

Data Preprocessing and Visualization

BeeNeural Intern Portal — Data Science & Analytics Task

Tools: Python, Pandas, NumPy, Matplotlib

Dataset: Student Performance Dataset (15 records, 7 columns)

1. Introduction

This project demonstrates a basic data science workflow using Python. The dataset was inspected, cleaned, analyzed, and visualized using Pandas, NumPy, and Matplotlib. The objective was to practice data preprocessing, exploratory data analysis (EDA), and data visualization.

2. Dataset Overview

Column	Description
Name	Student name
Age	Student age
Gender	Student gender
Study_Hours	Study hours per day
Attendance	Attendance percentage
Math_Score	Math examination score
English_Score	English examination score

3. Data Preprocessing

The dataset structure, column names, data types, missing values, and duplicate records were checked. For preprocessing practice, three numerical values were temporarily set to missing. Missing numerical values were then replaced using the median of their respective columns. Duplicate records were checked and removed. The final cleaned dataset contained no missing values and no duplicate rows.

Check	Result after cleaning
Missing values	0
Duplicate rows	0
Final rows	15
Final columns	7

4. Exploratory Data Analysis

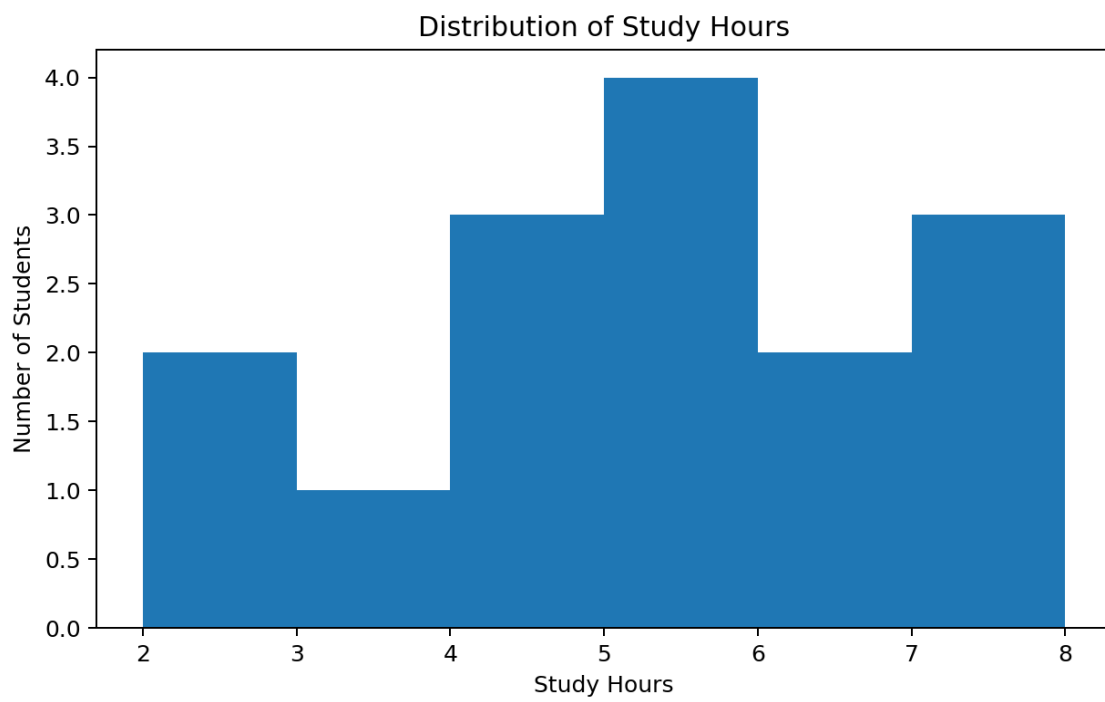
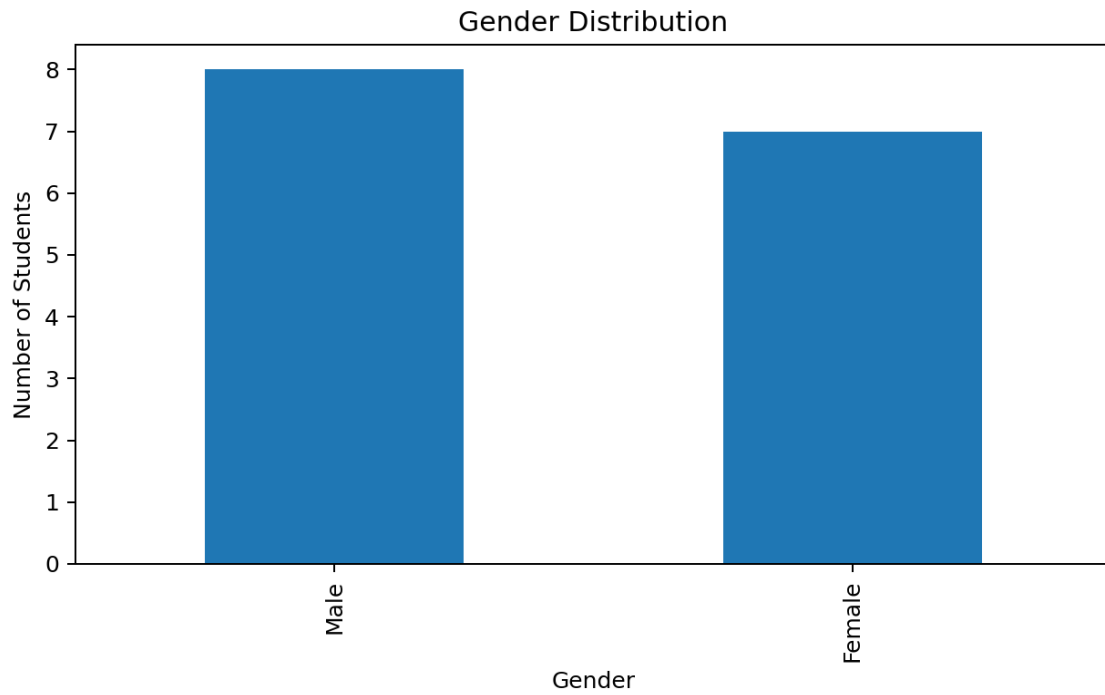
The average study time was 4.87 hours, while average attendance was 85.33%. The average Math score was 77.17, and the average English score was 79.20. The highest Math score was 95, and the highest English score was 94.

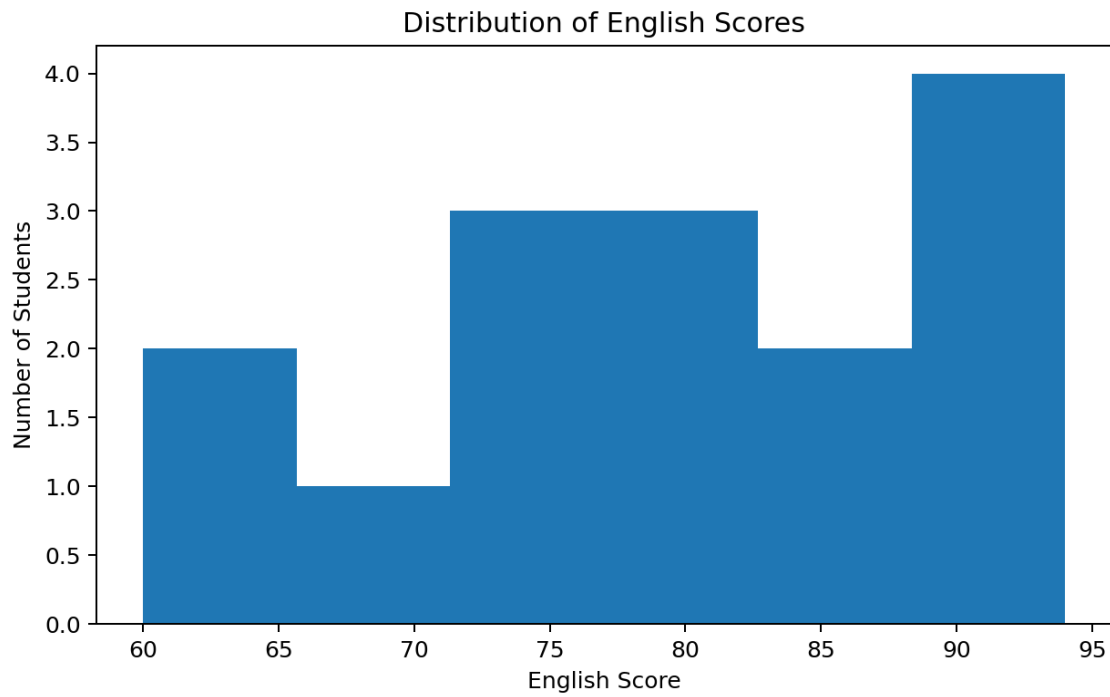
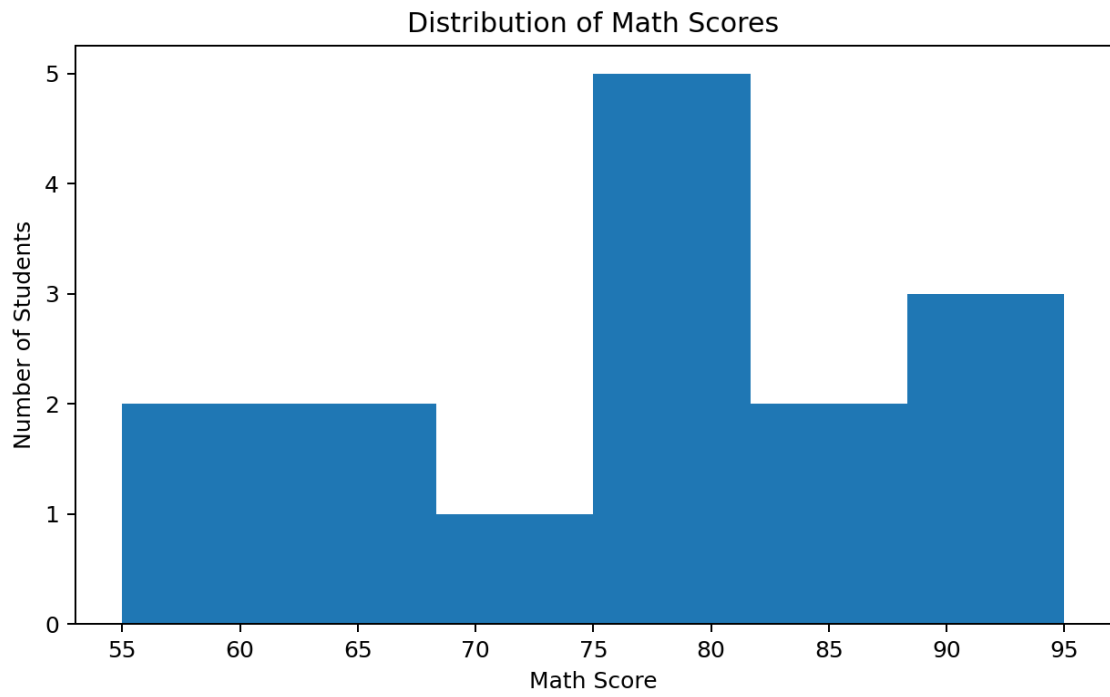
5. Correlation Analysis

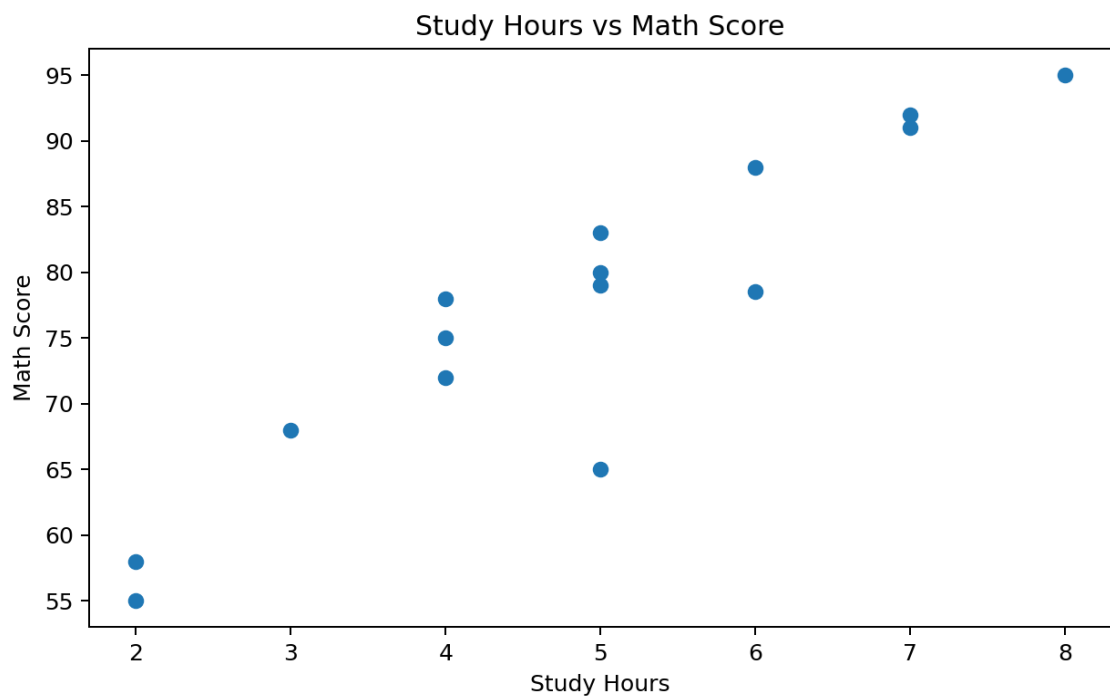
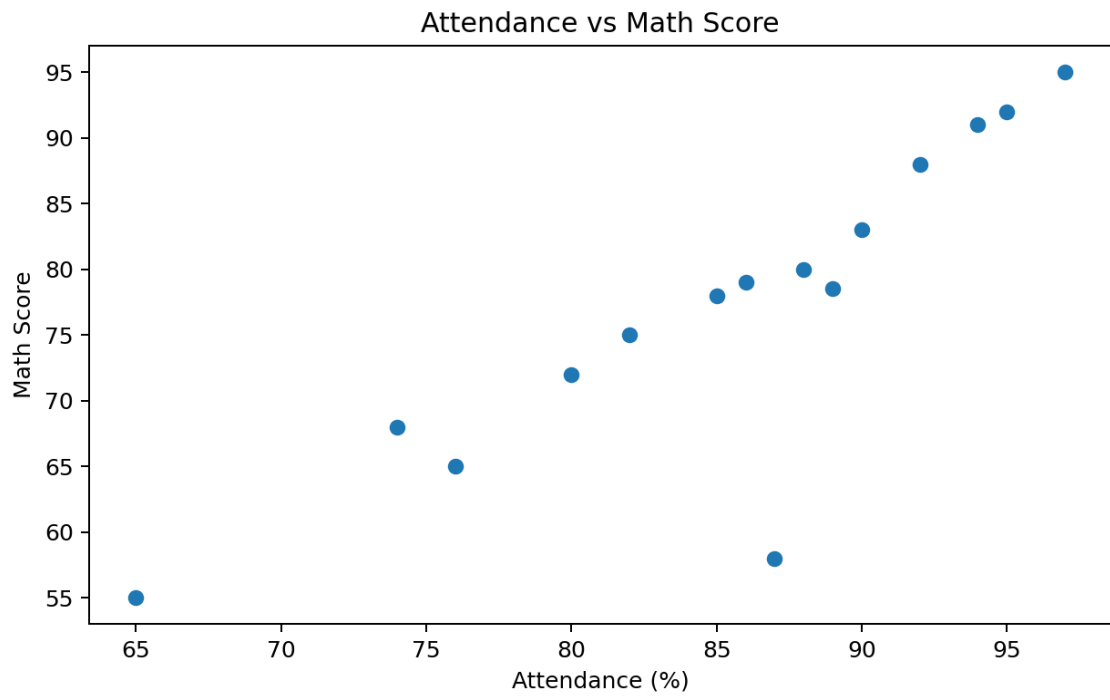
Variable	Correlation with Math Score
English Score	0.9258
Study Hours	0.9153
Attendance	0.8571

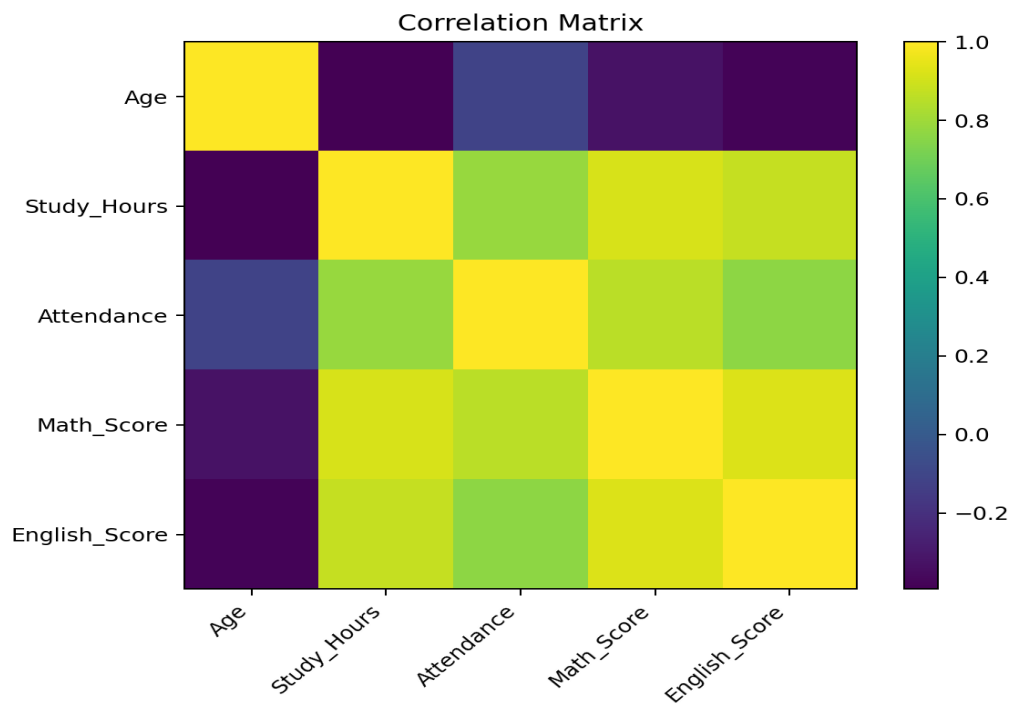
The correlations are descriptive relationships within this dataset and should not be interpreted as proof of causation.

6. Visualizations









7. Conclusion

The project completed the requested data preprocessing, exploratory analysis, and visualization workflow. The cleaned dataset was saved as a CSV file, and multiple visualizations were created to examine distributions and relationships among study habits, attendance, and academic scores. The analysis also demonstrates the use of Pandas for data manipulation, NumPy for numerical operations, and Matplotlib for visualization.

8. Project Files

The submission package includes the cleaned dataset and chart images. The Python notebook created in Google Colab should be uploaded to the GitHub repository together with the README file and these outputs.